Student's Full Name: ……

THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM TWO SECOND PRE–MOCK EXAMINATIONS

031 BIOLOGY

Time: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** questions in **all** sections.
3. Section A carries **fifteen (15)** marks, section B **seventy (70)** marks and section C carries **fifteen (15)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Name in Full** on the top right hand corner of every page.

FOR EXAMINER'S USE ONLY		
QUESTION NUMBER	**SCORE**	**EXAMINER'S INITIALS**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

K.S.S BIO PRE MOCK2 2025

SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the space provided.

(i) Which of the following food substances is chemically digested in the mouth?

A Starch B Protein
C Lipid D Glucose

(ii) The method of waste disposal that involves burning hazardous wastes at very high temperature is known as:

A Recycling B Landfill
C Incineration D Reducing

(iii) Which of the following plants lack xylem and phloem?

A Mushrooms and mosses B Club mosses and horsetails
C Ferns and mosses D Mosses and liverworts

(iv) Which of the following statements is correct regarding eukaryotic cells?

A They lack organelles like ribosomes and Golgi apparatus
B They are found in plants only
C They have nucleus with a nuclear envelope
D They do not have true nucleus

(v) Conditions or factors that can change or be changed in an experiment are called:

A Hypothesis B Variables
C Problems D Conclusions

(vi) All of the following are commonly observed in adolescent girls, **EXCEPT:**

A Development of soft voice
B Rapid increase in body size and height
C Onset of wet dreams and production of sperms
D Onset of ovulation and menstruation

(vii) Incomplete treatment lowers the body immunity because:

A All pathogens are completely destroyed
B Pathogens are not completely destroyed
C Antibodies are produced by pathogens
D White blood cells accumulates in the body

(viii) We are insisted to abstain from unprotected sexual intercourse so as to avoid different diseases such as:

A Tuberculosis and heart failure B Trichomoniasis and chlamydia
C Cholera and gastric ulcers D Rickets and gonorrhea

(ix) By which process do plants get their food?

A Respiration B Digestion
C Photosynthesis D Phototropism

(x) A **coconut plant** is scientifically known as ***Cocos nucifera.*** This means that it belongs to:

A Species plantae and division bryophyta
B Division filicinophyta and genus nucifera
C Species nucifera and genus cocos
D Genus nucifera and species cocos

ANSWERS

QUESTION	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ANSWER										

2. Match the characteristics of living things in **List A** with their correct definitions in **List B** by writing its letter below the number of the corresponding item number in the space provided.

LIST A	**LIST B**
(i) Growth	A The act of changing position
(ii) Movement	B Removal of metabolic wastes from the body
(iii) Nutrition	C Permanent increase is size and dry mass of an organism
(iv) Excretion	D Breaking down of food in living cells to produce energy
(v) Respiration	E The process by which an organism obtains food from the environment necessary for its survival
	F The act of responding to the stimuli
	G Giving rise to new ones of the same kind

ANSWERS

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B					

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) Biology is not only about the study of plants and animals. It also deals with the study of many other organisms. Outline three examples of organisms studied in each of the following fields.

(i) Entomology

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(ii) Microbiology

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(b) Draw the diagram of the following apparatii and state how each is used in studying living things.

S/N	APPARATUS	DIAGRAM	USES
(i)	Fish (Scoop net)		
(ii)	Pooter		
(iii)	Mortar and pestle		

4. An ecologist carried out a survey to observe different components of the environment present in a certain ecosystem. The results of his survey showed that there were grasses, eagles, fertile soil, rabbits, mushrooms and spring water.

(a) What do you understand by the term ecosystem? ..……

..…….

..…….

(b) Identify all abiotic components present in the ecosystem above.

..…….

..…….

(c) Which of the above organisms are termed as:

(i) Producers ……

(ii) Organisms of the third trophic level. ……

(iii) Decomposers? …….

(d) Draw a food chain for all living organisms found in the ecosystem above.

..…….

(e) What would be the effect to the population of grasses and rabbits if all eagles are eliminated from the ecosystem above?

..…….

..…….

5. (a) With examples, explain any two types of diseases based on their mode of transmission.

(i) ……

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(ii) ……

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(b) As a form two biology student, describe in four points how you will protect your family members at home from being bitten by mosquitoes.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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6. (a) What is blood? …….

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(b) What role does the following components of blood play in the body?

(i) Red blood cells (Erythrocytes)

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(ii) White blood cell (Leukocytes)

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(iii) Platelets (Thrombocytes)

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(c) In which ways is transpiration necessary for plants? Provide three points.

(i) ……

(ii) ……

(iii) ……

7. Penina cooked rice and put it into a pot. She forgot to eat the rice and it remained in the pot for two weeks. Later, she opened the pot and found that the rice was covered by a mass of grayish structures which look like threads.
   (a) Give the common name of the structures that were observed on rice. ……
   (b) State the kingdom and phylum of the structures named in (a) above.

   ..…….

   ..…….
   (c) What is the mode of nutrition of these structures? …….
   (d) Outline three key differences between plants and the structures grown on rice.

S/N	PLANTS	STRUCTURES ON RICE
(i)		
(ii)		
(iii)		

   (e) Draw the diagram of the structures described in this question and label three parts.

8. (a) With regard to food test procedure, give the reasons for each of the following observations:
   (i) Non-reducing sugar is boiled first with hydrochloric acid during food test.

   ..……

   ..……
   (ii) A series of colour change that ends at brick red colour is observed when reducing sugars are boiled with Benedict's solution.

   ..……

   ..……

(b) Paulina Loibanguti weighs 59kilograms on a beam balance and she measures 147centimetres in height.

(i) What is her Body Mass Index (BMI)?

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(ii) According to the World Health Organization, what is the health status of Paulina based on her BMI?

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(iii) State three health problems that Paulina is likely to encounter due to her BMI

..……

..……

..……

(iv) As a young biologist, provide a professional advice to Paulina which corresponds to her health status. Give two points.

..……

..……

9. Zainabu and her friends were playing in the ground. Unfortunately, she fell down and got injured at her knee where she started bleeding. Her friends Naini and Lusia put on gloves and helped her by washing her injuries, applying antibiotics and covering them with a plaster.

(a) (i) Give the name of the action done by Naini and Lusia to their friend Zainabu.

..……

(ii) Define the term named in a(i) above. ..……

..……

..……

(b) Give the name given to Naini and Lusia according to the part they played in such incidence.

……..……

(c) Why was it important for them to wear gloves before washing the injuries of Zainabu? Give two points.

……..……

……..……

(d) Why did they apply antibiotics to the injuries? ..……..

……..……

(e) What would happen to Zainabu if her friends would not help her? Provide four points.

(i) ..……

(ii) ..……

(iii) ..……

(iv) ..……

SECTION C (15 Marks)

Answer the question **ten (10)** below

10. Assume that you are a competent virologist and you have been invited as a guest speaker to teach form one students about this branch of biology. Prepare your hints for presentation basing on the advantages and disadvantages of viruses in the environment. Give three points in each case.

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